formats. However, we find that this approach may hinder the model's exploration and ultimately harm its performance particularly for the base models which struggle with following the format in the initial stage, as detailed in §3.1.

### B.3 Models

We conduct zero RL training experiments on Llama-3.1-8B (Dubey et al., 2024), DeepSeek-Math-7B (Shao et al., 2024), Mistral-v0.1-7B (Jiang et al., 2023), Mistral-Small-24b-Base-2501 (Mistral AI, 2025), and Qwen-2.5 (0.5B, 1.5B, 7B, 14B, 32B) (Yang et al., 2024a). As we perform experiments for a variety of models, under extremely simple settings with small, simple datasets and only correctness reward, we refer to our obtained models as *SimpleRL-Zoo* to represent a simple training recipe for a zoo of open base models. For models with weaker instruction-following capabilities (Llama-3.1-8B, Mistral-v0.1-7B, and Qwen-2.5-0.5B/1.5B), we employ simpler prompts (Chern et al., 2023) requiring only step-by-step reasoning. For models with stronger instruction-following abilities, we use more complex prompts (Yang et al., 2024a) that require the final answers to be placed in boxes. In our preliminary experiments, we observe that using complex prompts with models that have weak instruction-following capabilities often results in large amounts of irrelevant or nonsensical content being generated early in training, leading to instability. The content of simpler prompts and more complex prompts is shown in Figure 10 in Appendix.

### B.4 Benchmark

We evaluate performance on standard mathematical reasoning benchmarks, including GSM8K (Cobbe et al., 2021), MATH 500 (Hendrycks et al., 2021), Minerva Math (Lewkowycz et al., 2022), and OlympiadBench (He et al., 2024), as well as on competition-level benchmarks such as AIME 2024 and AMC 2023.

### B.5 Training and Evaluation Details

We train our models using the verl (Sheng et al., 2024) framework. And we typically use the same set of hyperparameters to train and evaluate all models in the SimpleRL-Zoo series in default main experiment setting. We use a prompt batch size of 1,024 and generate 8 rollouts per prompt, with a maximum rollout length of 8,192 tokens. Training is performed using a mini-batch size of 256. The default sampling temperature is set to 1.0, and the clip ratio is 0.2. For models ranging from 0.5B to 14B parameters, we use a KL loss coefficient of 1e-4. For models larger than 14B, the KL loss coefficient is set to 1e-3. We build our evaluation script based on Yang et al. (2024b), using a temperature of 1.0 and a maximum generation length of 16K tokens. To ensure consistency, we adopt the same prompt template used during training. For most benchmarks, we report pass@1 results. However, for AIME 2024, which contains fewer problems, we report both pass@1 and average accuracy (avg@32), computed over 32 generated samples per problem.

## C Detailed Evaluation Metrics

**Reasoning Behavior Ratio:** To better understand the model's reasoning patterns throughout the training process, we adopt the cognitive behavior framework proposed by Gandhi et al. (2025) and use GPT-4o (Hurst et al., 2024) to identify reasoning-related behaviors, including "Backtracking", "Verification", "Subgoal Setting", and "Enumeration". We report the ratio of responses that contain such cognitive behaviors. While some recent studies suggest tracking reflection behavior using related keywords (Yeo et al., 2025; Xie et al., 2025) as monitoring signals, we argue that these keywords only exhibit only a weak correlation with high-level reasoning patterns like reflection and verification. As a result, they fail to adequately capture the development of these reasoning processes. Further details can be found in Appendix I.

**Clip Ratio:** In the early stages of training, the base model exhibits weak instruction-following ability and often fails to stop appropriately, resulting in irrelevant or excessively

**Simple Prompt**

Question:
{input}
Answer:
Let's think step by step.

**Complex Prompt**

<|im_start|>system
You are a helpful assistant.<|im_end|>
<|im_start|>user
{input}
Please reason step by step, and put your final answer
within \\boxed{}.<|im_end|>
<|im_start|>assistant
{output}

Figure 10: Comparison between simple prompts and more complex prompts.

long outputs. After training collapses, the model may also generate repetitive or overly extended responses. Since the model has a fixed maximum context length, such outputs may be truncated during both training and evaluation. To monitor this issue, we define the proportion of truncated outputs as the "Clip Ratio".

**Average Stopped Length:** Generations that are truncated often result from issues such as repetitive patterns or incomplete reasoning, which typically do not contribute to effective trajectories. To account for this factor, we introduce a new metric to track the average length of responses that are stopped under normal conditions. It is a more reliable metric to consider only valid responses, thereby eliminating the interference caused by unstopped responses.

**Pass@k Accuracy:** We track the pass@k accuracy, which represents the percentage of questions for which at least one correct response is obtained when sampling k responses per question. Pass@k serves as an indicator of the model's exploration capabilities and is particularly relevant for RL, as it reflects the model's ability to generate responses that can achieve a positive reward. Previously, some researchers believed that RL training might merely reorder responses within the original model distribution, as evidenced by the lack of improvement in pass@k accuracy following RL training (Shao et al., 2024).

## D Detailed Result of SimpleRL

Following the setup described in Section 2.1, we perform "zero training" on various base models. The trained models are then evaluated on multiple benchmarks, including GSM8K, MATH 500, Minerva Math, OlympiadBench, AIME2024, and AMC2023. The average results across all these benchmarks are presented in Figures 1 and 4. In this section, we provide more detailed results. Figure 11 illustrates the trends in accuracy and response length, while Figure 12 shows the trends in clip ratio and stopped length.

## E Quantitative Behavior Validation

We assess the consistency between GPT-4o labeled reasoning behaviors and human annotations by having human experts annotate 105 model outputs. Table 3 below presents the prediction rates and agreement rate. The prediction rate reflects how frequently each reasoning behavior is identified, while the agreement rate is the proportion of data on which the labelers (Human and GPT-4o) make the same prediction.

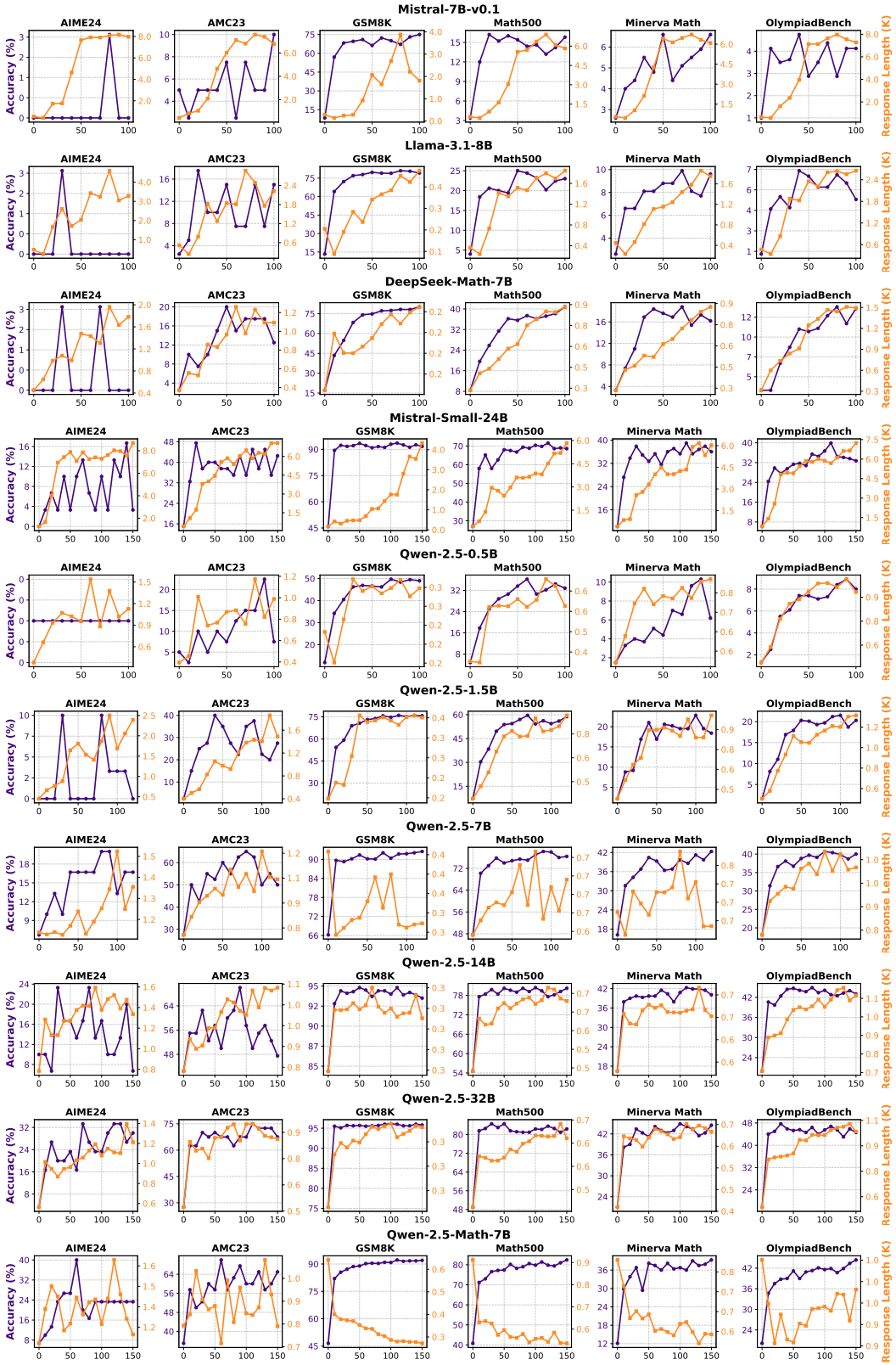


Figure 11: A detailed evaluation of accuracy and response length throughout the training steps for various models. The x-axis represents the training steps, with the purple line showing the accuracy trend and the yellow line depicting the response length.